FARM MANAGEMENT SYSTEM

Short Project Documentation

Platform: Salesforce

Project Type: Group Project – Five Members

1. Introduction

The Farm Management System (FMS) is a Salesforce-based application developed to organize and manage agricultural information in a centralized system. The project provides separate modules for Farms, Crops, Farmers and Buyers, together with reports and a dashboard for structured data viewing and analysis.

2. User Story

As a farmer, farm manager or system administrator, the user needs a centralized platform to record and manage farm, crop, farmer and buyer information securely and systematically. The system supports organized data entry, validation, reporting, dashboard visualization and controlled access to improve efficiency and reduce manual work.

3. Objectives

• Centralize agricultural information in Salesforce.

• Manage farmer, farm, crop and buyer data using dedicated objects.

• Improve data accuracy through picklists and field dependencies.

• Provide reports for structured data analysis.

• Provide a dashboard for quick visual understanding of available information.

• Support role/profile-based access and record sharing.

4. Salesforce Environment Setup

A Salesforce Developer Org was created using the Salesforce Developer signup process. The account was verified through email, a password and security question were configured, and the Salesforce Setup environment was accessed.

5. Custom Objects

Object	Record Name	Purpose
Farm	Farm Name	Stores farm-related information
Crop	Crop Name	Stores crop and production information
Farmer	Farmer Name	Stores farmer information
Buyer	Buyer Name	Stores buyer information

6. Custom Tabs and Application

Custom tabs were created for Farm, Crop, Farmer and Buyer. A Lightning App named **Farm Management System** was then created with Farm, Crop, Farmer, Buyer, Reports and Dashboards in its navigation menu. The System Administrator profile was assigned to the application.

7. Custom Fields and Relationships

Farm: Farm Type (Crop, Livestock, Mixed), Soil Quality (Organic, Cover Crops, Composting), Irrigation Type (Sprinkler irrigation, Drip irrigation, Localized irrigation), Location, Size and a Master-Detail relationship with Farmer.

Crop: Crop Name, Planting Date, Harvest Date, Lookup relationship with Farm, Actual Yield, Total Acres Harvested and a formula field Expected Yield per Acre.

Farmer: Contact Number, State, District and Village. A field dependency was created with State as the controlling field and District as the dependent field.

Buyer: Contact, Email, Address and Buyer Type (Wholesale, Retail).

8. Validation and Data Dependency

The Farmer object uses a State–District field dependency. When a State is selected, the corresponding District values can be displayed. This helps reduce incorrect data entry and improves consistency.

The Crop object also contains a formula field that calculates Expected Yield per Acre from Actual Yield and Total Acres Harvested.

9. Security and Access Management

Custom profiles were created for Sales Department and Farm Manager by cloning the Standard User profile. A Farm Manager role was also created. A Farm Manager user was configured with the Standard Platform User license, Farm Manager role and Farm Manager profile.

Sharing rules were configured to support controlled record access. Crop-related records were shared with the Farm Manager role with Read/Write access, and Buyer records were shared with the Sales Department with Read/Write access.

10. Records

A Farmer record was created through the Farm Management System application. The project demonstration includes a farmer record named **Ravi Kumar**. At the current demonstration stage, Farms, Crops and Buyers lists may display no records; the corresponding modules remain available for creating and managing records.

11. Reports

Report	Purpose
Farmers with Farms	Displays farmer and farm-related information.
New Buyers	Displays buyer-related information.
New Crops	Displays crop-related information.

The Farmers with Farms report was created using the required fields and grouping options and saved in a public folder. Reports provide a structured way to view available information.

12. Dashboard

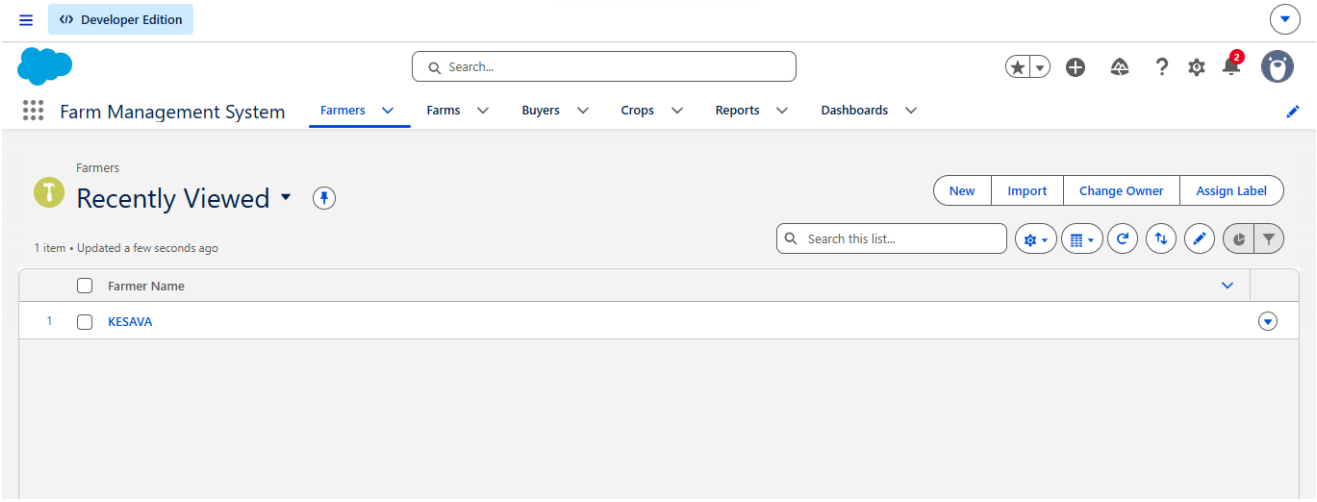
A dashboard named **Farmers with Farms** was created using the Farmers with Farms report. A Gauge component was added to provide a visual representation of the report information. The dashboard can be accessed from the Dashboard tab of the Farm Management System application.

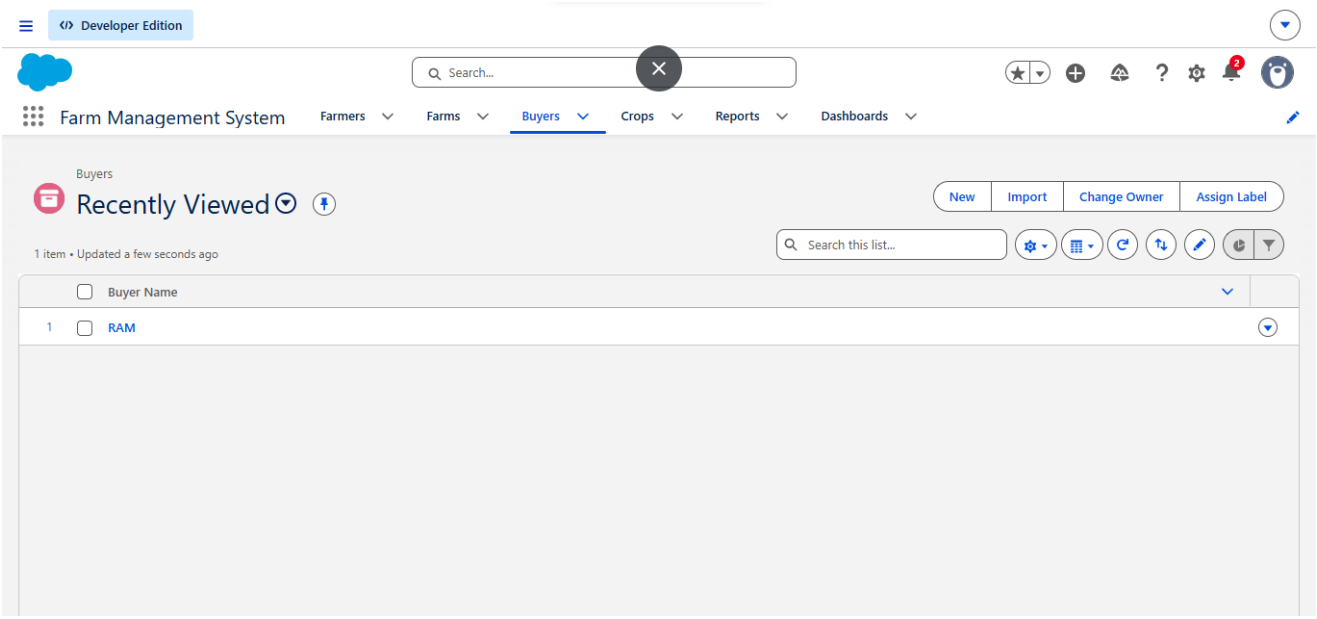
13. Project Workflow

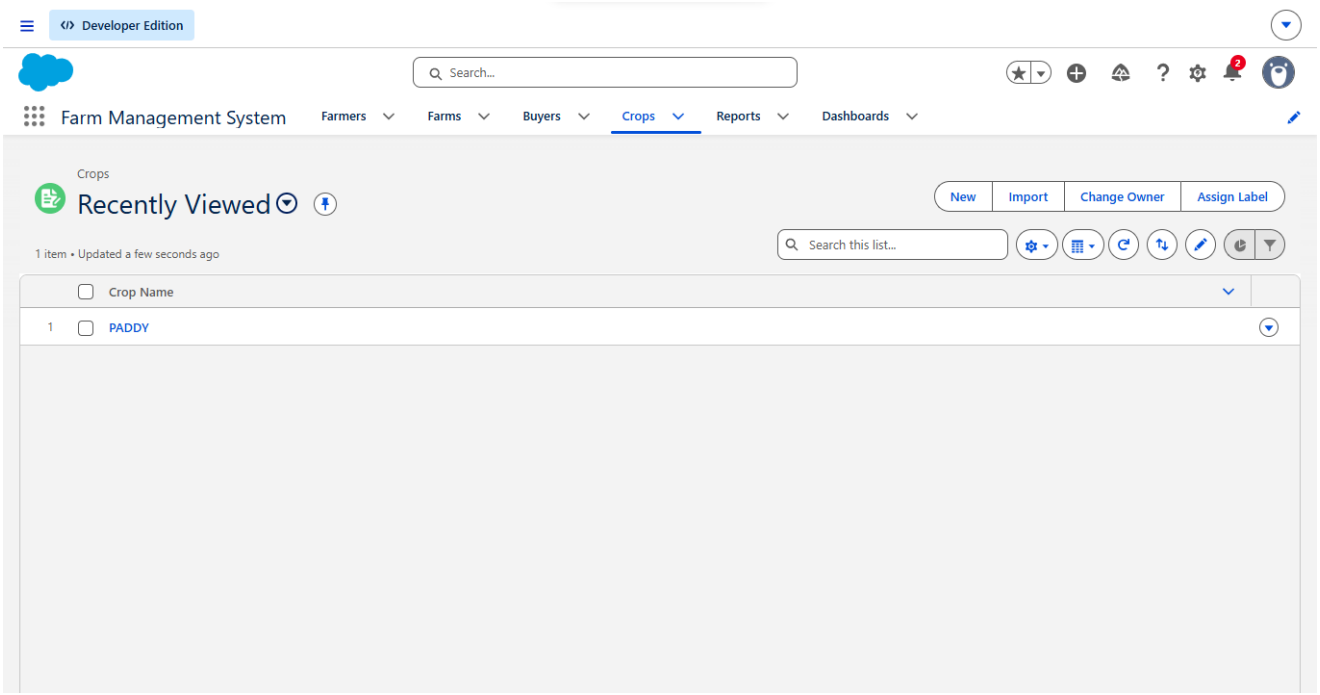
Developer Org → Custom Objects → Custom Tabs → Lightning App → Custom Fields & Relationships → Profiles, Roles & Users → Sharing Rules → Records → Reports → Dashboard

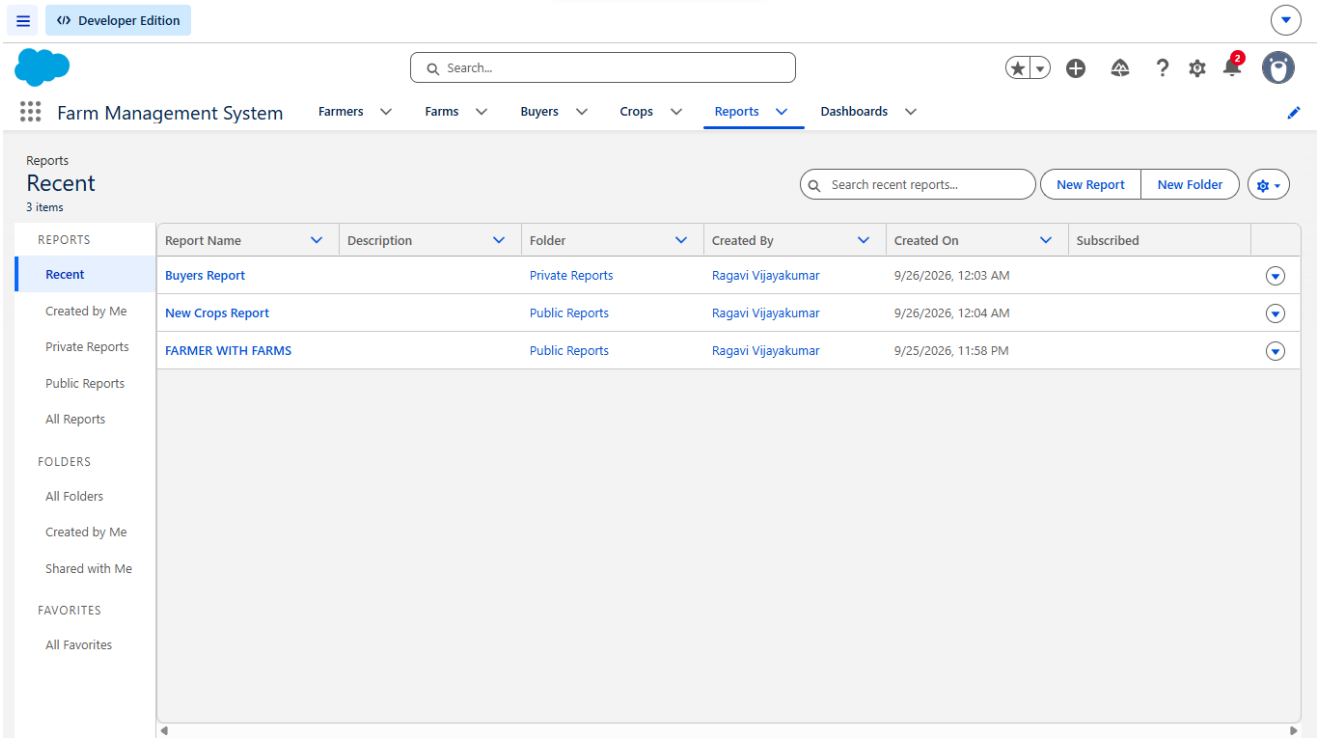
14. Screenshot

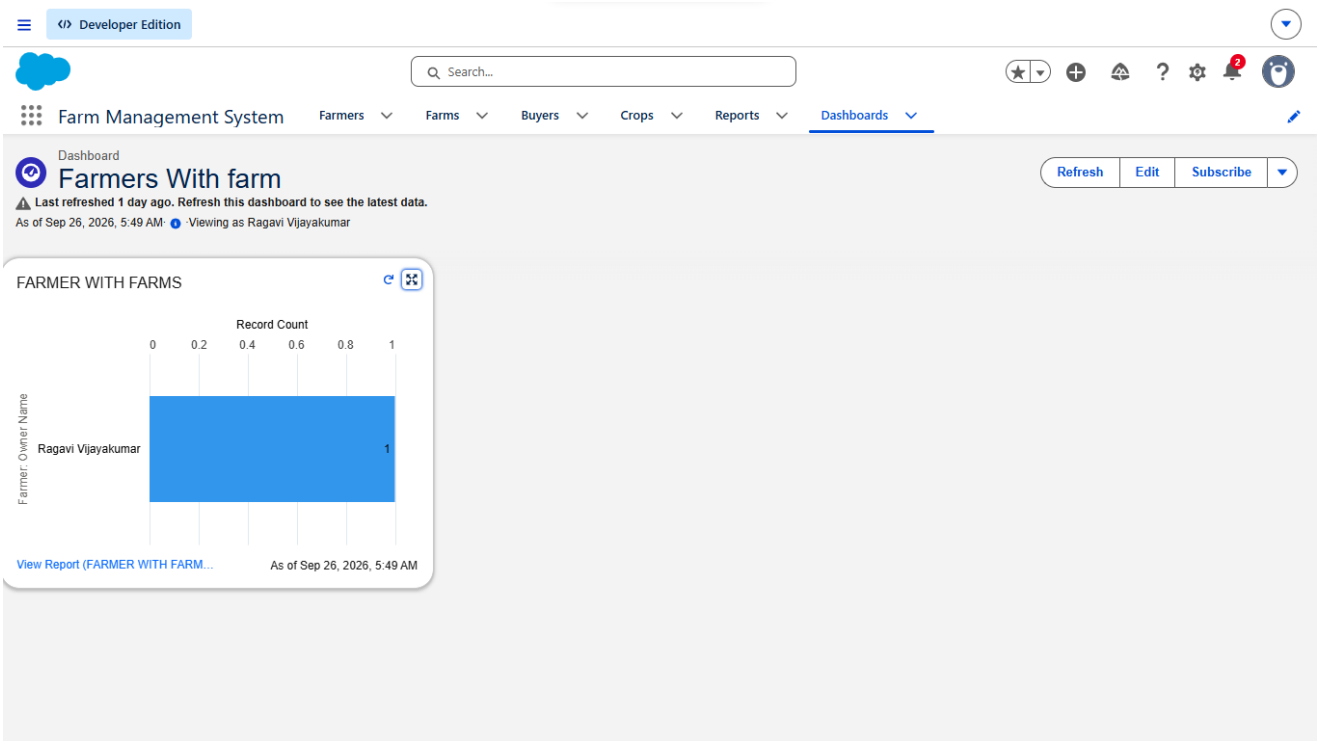
Farm Management System






Developer Edition
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Farms
Buyers
Crops
Reports
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Crops
Recently Viewed
New
Import
Change Owner
Assign Label
1 item • Updated a few seconds ago
Search this list...
Crop Name
1
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Buyers
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Reports
Dashboards
Reports
Recent
3 items
Search recent reports...
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REPORTS
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15. Conclusion

The Farm Management System demonstrates how Salesforce can be used to organize agricultural information through custom objects, fields, relationships, security settings, reports and dashboards. The system provides a structured foundation for managing farmer, farm, crop and buyer information and supports better access to organized data.

16. Future Enhancements

• Add more farm, crop and buyer records for larger-scale testing.

• Add additional automation and validation rules.

• Expand dashboards with more analytical components.

• Introduce notifications and workflow automation.

• Add more detailed sales and productivity analysis.